

DevSecOps:

Building Secure Software with Open Source Tools

Security needs to be embedded in the design of all modern software products. This is where DevSecOps and its toolchain play a significant role. Find out how they help, and what are the best practices for implementing this toolchain.



In today's fast-paced digital economy, the speed at which software can be developed and delivered has become a major competitive advantage. The adoption of DevOps practices has transformed software delivery by fostering close collaboration between development and operations teams, and by relying heavily on automation, continuous integration (CI), and continuous delivery (CD) pipelines. These

practices enable organisations to release features, updates, and patches far more quickly than in the past. However, this increased velocity can come at a cost. If security is treated as an afterthought — checked only at the end of the development process — vulnerabilities can go unnoticed until it is too late. In an era where cyberattacks are increasingly sophisticated, this lag in addressing security can lead to severe consequences such as data

breaches, regulatory non-compliance, and reputational damage.

This is the gap that DevSecOps aims to fill. DevSecOps builds upon the DevOps philosophy but integrates security into every stage of the software development lifecycle (SDLC). The guiding principle here is 'shifting security left', which means incorporating security practices — such as vulnerability scanning, threat modelling, and compliance checks